CARDINAL SURVEYS COMPANY

INJECTION PROFILE W/CALIPER

Company CHEVRON U.S.A., INCORPORATED
Well J.T. MCELROY UNIT NO. 1606W
Field MCELROY
County CRANE State TEXAS
Location 2317' FSL & 2498' FEL SEC. 199, BLK. F, CCSD & RGNG SURVEY API NO: 42-103-35811
File Number 20,782

Company Chevron U.S.A., Incorporated
Well J.T. McElroy Unit No. 1606W
Field McElroy
County Crane
State Texas

Date	5-07-12				
Depth Driller	2763'				
Plug Back					
Depth Logger	2761'				
Bottom Logged Interval	2761'				
Top Log Interval	2400'				
Recorded By	NEWMAN				
Witnessed By	T. JONES				
Base Location	ODESSA, TX				
Unit No.	8727				
Equipment Operator	PELLISSIER				
Type Of Well	INJECTION				
Status	INJECTING				
Type Of Fluid	WATER				
Fluid Level	FULL				
Rate	177 BPD				
Pressure	380-550 PSI				

Permanent Datum G.L. Elevation 2552'
Log Measured From K.B. 12' Above Perm. Datum
Drilling Measured From K.B.

Elevation
K.B. 2564'
D.F. 2563'
G.L. 2552'

Borehole Record

Run Number	Bit	From	To

Tubing Record

Size	Weight	From	To
2 3/8"		SURFACE	2648'

Casing Record

	Size	Wgt/Ft	Top	Bottom
Surface String	8 5/8"	24#	SURFACE	825'
Intermed String				
Production String	5 1/2"	15.5#	SURFACE	2671'
Liner				

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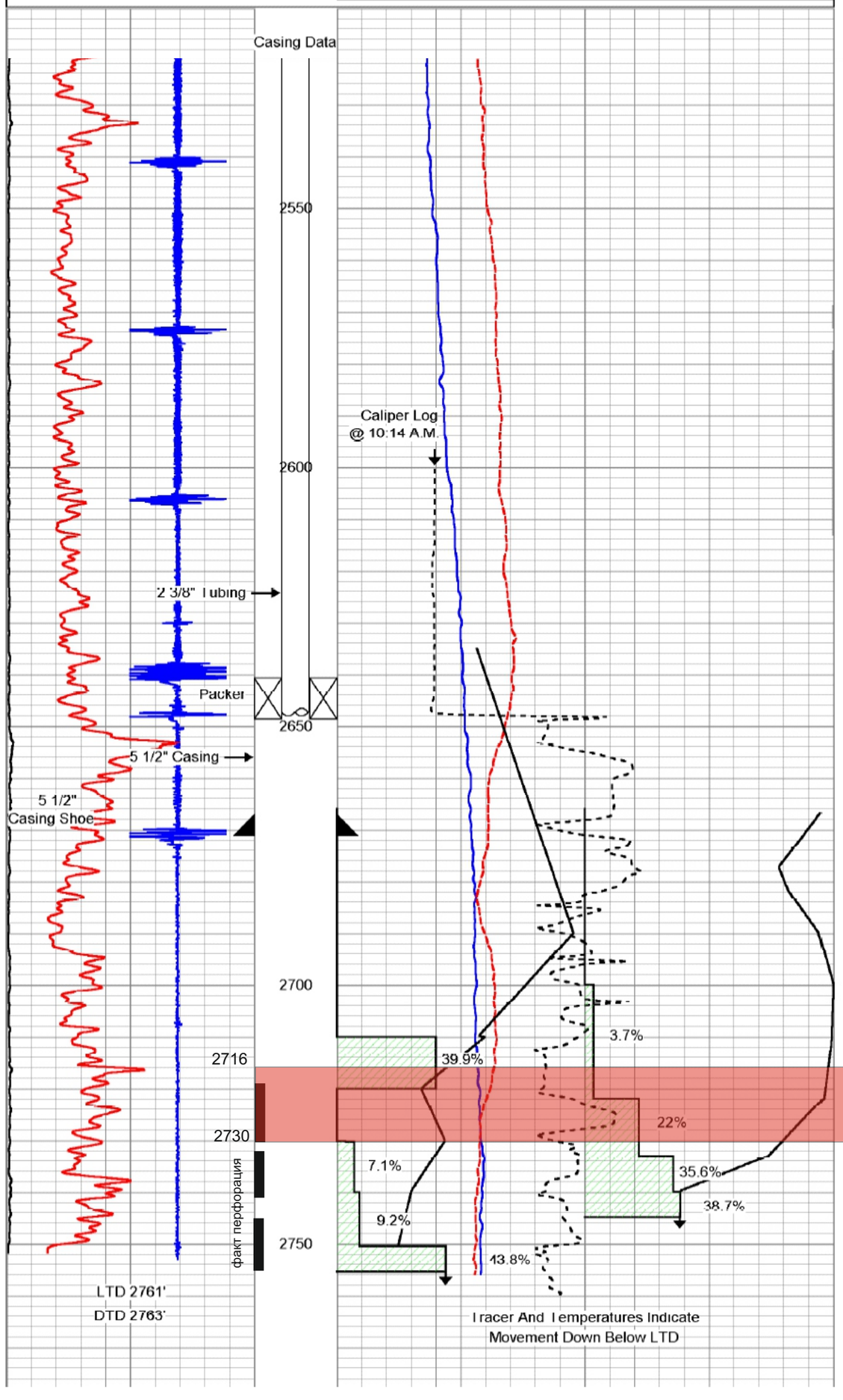
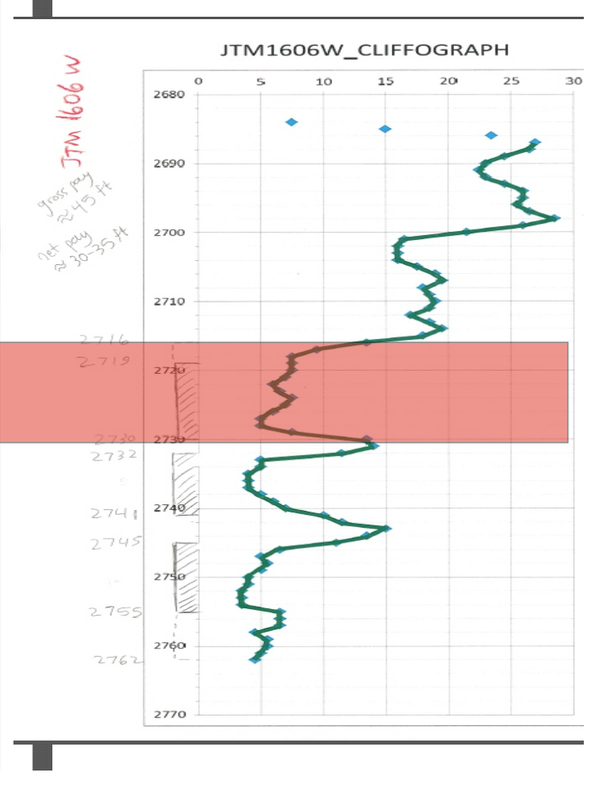
TOOL CONFIGURATION: CAL, TEMP, CCL, DET, EJECT. (1 1/4") SCINTILLATION DETECTOR
CORRELATED TO: TIED TO CASING SHOE @ 2671'
OPEN HOLE: 2671'-63'

CONCLUSION: Reported injection rate 177 BPD @ 380-550 PSI.
Used slug no. 13, 277.1 BPD, for 100% rate to calculate velocity profile.
Used run 1 for 100% area. Used runs 5 and 7-9 to plot intensity profile.
Tracer and temperatures indicate movement down below LTD. Slug no. 2.
No channel up from 5 1/2" casing shoe indicated. Slug no. 11.
No packer leak indicated. Slug no. 13.

Database File: sc59946.db
Dataset Pathname: MERGE1
Presentation Format: SC59946
Dataset Creation: Mon May 07 10:23:09 2012
Charted by: Depth in Feet scaled 1:240

-9	Collar Log	4	82	Injection Temperature (degF)		87
-10	Base Gamma Ray, CPS	1500	82	1 Hr. Shut In Temperature (degF)		87
-10	Depth Control Log, CPS	50	0	Caliper Log		10
			0	Velocity Loss, %	100	0 Intensity Loss, %	100
			0	Velocity Flow, (b/d)	490	0 Intensity Flow,% (%)	100